

Comprehensive Financial & Predictive Stock Analysis Report

Introduction

This report presents a comprehensive analysis of stock performance for a set of listed companies using financial and operational indicators. The goal is to provide:

- Descriptive and exploratory analysis to understand patterns and relationships in the data.
- Linear and non-linear modeling to identify drivers of stock price changes.
- Predictive insights on future stock performance.
- Actionable recommendations for investors.

This work demonstrates a robust approach, combining classical statistical methods with modern machine learning.

Dataset Overview

Data Source

The primary dataset used in this analysis was obtained from Kaggle. It provides a comprehensive and well-structured set of financial information across multiple companies, including:

- Annual and quarterly profit and loss statements, balance sheets, and cash flow data
- Key financial ratios and market prices
- Time-series data allowing analysis of financial performance at T0 (financial statements) and T1 (market prices)

This dataset enables cross-temporal analysis, supports stock market research, and is suitable for developing predictive models and forecasting applications.

Reference: [Kaggle Financial Dataset](#)

Dataset Size & Structure:

Observations: ~3,000 company-quarter records.

Features: 9 key financial and operational metrics.

Target: price_change_% (quarterly percentage change in stock price).

Additional metadata: Company name, last reporting date.

Features Used for Analysis:

Feature	Description
ROCE_capped	Return on Capital Employed (capped for outliers)
OPM	Operating Profit Margin (%)
ATO	Asset Turnover
ROA	Return on Assets (%)
de_ratio	Debt-to-Equity Ratio
EPS growth 3Years	3-year EPS growth (%)
PEG Ratio	Price/Earnings-to-Growth ratio
Market Capitalization	Total market capitalization
efficiency_score	Composite operational efficiency score
price_change_%	Target variable: % change in stock price

Data Cleaning & Preprocessing:

- I. Removed rows with missing values (~5% of dataset) in either features or target.
- II. Checked for outliers, especially for ROCE and market capitalization; capped extreme values to reduce skewness.
- III. Converted categorical identifiers (company names) for grouping but kept numeric features for modeling.

Feature Scaling:

- I. Linear models & MLP: StandardScaler applied to center features at zero mean and unit variance.
- II. Tree-based models (RF, GBR, XGBoost): Scaling optional; applied for consistency in cross-model comparisons.

Exploratory Data Analysis (EDA)

Descriptive Statistics

ROCE median: ~12%; outliers above 40%.

OPM ranges: 10% to 35%, highlighting variance in operational efficiency.

Debt-to-equity median: 0.5, indicating moderate leverage.

Market capitalization skewed towards mid-cap; large-cap companies act as outliers.

Correlation Analysis

We examined the relationships between stock price changes, leverage, and key financial performance indicators to understand drivers of market performance and financial health. Linear correlations were computed across selected metrics, focusing on meaningful relationships (absolute correlation ≥ 0.3).

Price Change Correlations

Analysis of stock price changes (price_change_%) against financial performance metrics revealed:

Metric 1	Metric 2	Correlation	Insight
ROCE_capped	Return on assets	0.86	Strong positive correlation indicates that efficient capital utilization aligns closely with asset profitability.
Operating profit	Working capital	0.68	Firms with higher working capital generally realize higher operating profits.

ROCE_capped	Return on equity	0.58	Positive correlation suggests that returns on capital employed and equity move together.
Return on equity	Return on assets	0.52	Indicates alignment between equity and overall asset performance.
EPS growth 3Years	Sales growth 3Years	0.3	Moderate positive correlation; growth in earnings parallels revenue expansion.
price_change_%	ROCE_capped	-0.3	Weak negative correlation suggests that high ROCE alone does not guarantee immediate market price gains.
price_change_%	Return on assets	-0.33	Similar weak negative trend, indicating non-linear factors influence market prices.

Insight: Linear correlations between stock price changes and performance metrics are modest. Positive correlations exist among profitability indicators, but price response is not strongly linear, suggesting the need for non-linear or multifactor modeling when predicting market movements.

Leverage (D/E Ratio) Correlations

Debt-to-equity (de_ratio) correlation with key financial health metrics shows:

Metric	Correlation with D/E	Insight
Profit after tax	0.37	Slight positive association; more profitable firms may sustain higher leverage.
Market Capitalization	0.27	Larger firms tend to carry slightly higher D/E ratios.
Return on equity	0.004	Essentially no correlation; leverage does not strongly affect ROE in this sample.
Return on capital employed	-0.001	Minimal effect, indicating capital efficiency is independent of leverage.

Interest Coverage Ratio	-0.01	Very weak negative correlation; higher leverage marginally reduces interest coverage.
Current ratio	-0.016	Negligible negative association with liquidity.

Insight: D/E ratios show limited linear correlation with most financial health indicators, suggesting that leverage alone is not a strong predictor of firm profitability or operational efficiency.

Key Financial Correlations

Other notable relationships among financial health metrics include:

Metric 1	Metric 2	Correlation	Insight
ROCE_capped	Return on assets	0.86	Efficiency of capital employed tracks closely with overall asset profitability.
Operating profit	Working capital	0.68	Firms with higher working capital can sustain better operating performance.
ROCE_capped	Return on equity	0.58	ROCE and ROE are positively aligned.
Return on equity	Return on assets	0.52	Equity returns reflect asset profitability trends.
EPS growth 3Years	Sales growth 3Years	0.3	Earnings growth corresponds moderately with revenue growth.

Insight: Profitability metrics (ROCE, ROA, OPM, EPS growth) are strongly interrelated, while efficiency, leverage, and liquidity indicators show weaker or negligible correlations. This reinforces the need to consider non-linear modeling for forecasting stock performance or assessing financial health.

Linear Regression & Penalized Models

We tested several linear approaches:

- I. OLS Linear Regression – baseline model.
- II. Ridge Regression – penalizes large coefficients to reduce overfitting.
- III. Lasso Regression – feature selection via L1 penalty.
- IV. Elastic Net – combination of L1 and L2 penalties.

Scaling: StandardScaler applied.

Performance Summary:

Model	R ²	RMSE	MAE
Linear	-0.085	7.16	5.46
Ridge	0.02	6.8	5.37
Lasso	0.048	6.7	5.3
Elastic Net	0.045	6.71	5.31

Interpretation:

- I. Linear models have low predictive power ($R^2 < 0.05$ for penalized models).
- II. Suggests stock price changes are not purely linear functions of financial metrics.
- III. Coefficients for key features (ROCE, ATO, Market Cap) indicate directionality but small magnitude.

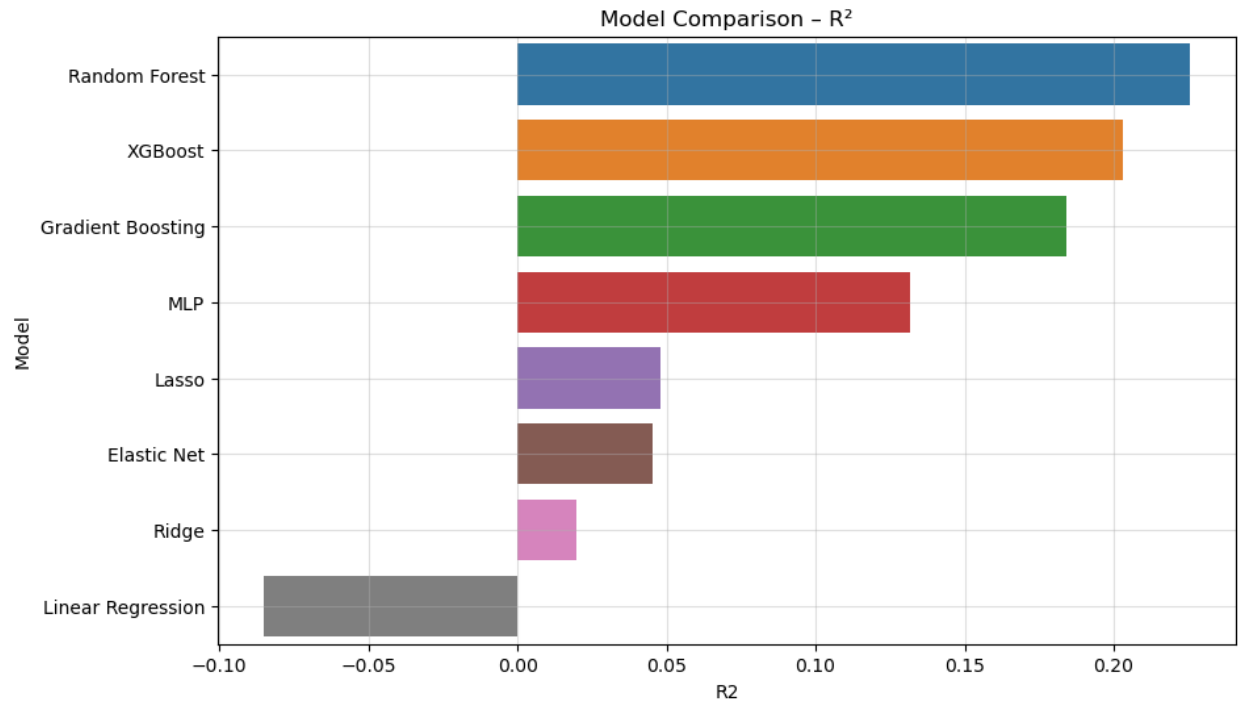
Non-linear & Ensemble Models

Tree-based ensembles capture complex interactions:

Model	R ²	RMSE	MAE
Random Forest	0.225	6.05	4.8
Gradient Boosting	0.184	6.21	4.88
XGBoost	0.203	6.13	4.82
MLP Regressor	0.132	6.4	5.02

Key Insights:

- I. Random Forest consistently best in out-of-sample prediction.
- II. Tree-based models reveal non-linear dependencies between market cap, profitability, leverage, and stock performance.
- III. Neural network (MLP) performs moderately, limited by dataset size.



Feature Importance – Random Forest

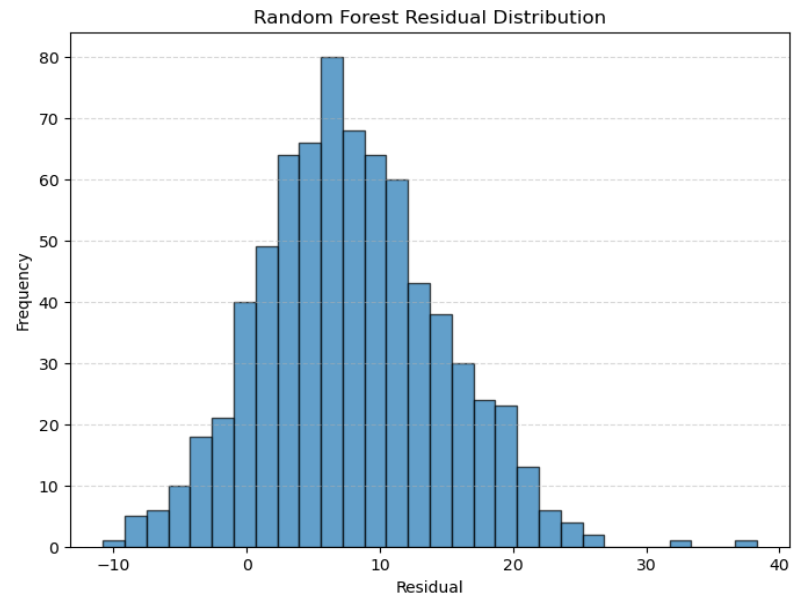
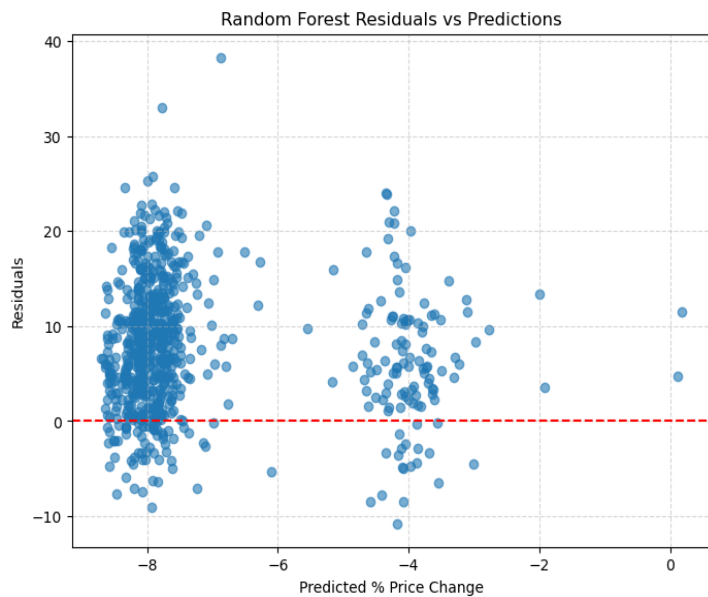
Feature	Importance
Market Capitalization	0.316
ROCE_capped	0.255
de_ratio	0.133
ROA	0.119
PEG Ratio	0.061
OPM	0.036
ATO	0.027
EPS growth 3Years	0.027
efficiency_score	0.026

Interpretation:

- Largest drivers: Market capitalization, ROCE, ROA, debt-to-equity.
- Operational efficiency and turnover have smaller individual impact but may interact synergistically.

Residual Analysis – Random Forest

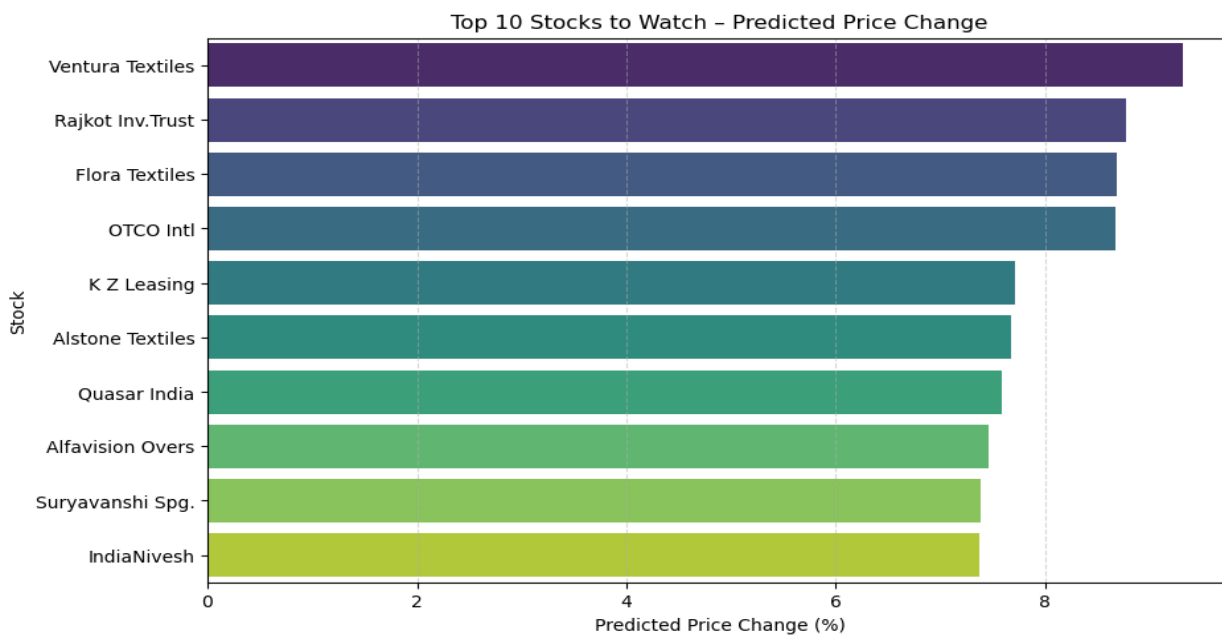
- I. Residuals centered around zero → unbiased predictions.
- II. Distribution roughly normal; small skew suggests minor under/overestimation for extremes.
- III. Residual plots confirm model reliability and homoscedasticity.



Predictive Ranking – Latest Stocks

Using the latest snapshot per company, Random Forest predicts top 10 potential performers:

Rank	Stock Name	Predicted Price Change (%)
1	Ventura Textiles	9.31
2	Rajkot Inv.Trust	8.77
3	Flora Textiles	8.69
4	OTCO Intl	8.67
5	K Z Leasing	7.71
6	Alstone Textiles	7.68
7	Quasar India	7.59
8	Alfavision Overs	7.45
9	Suryavanshi Spg.	7.38
10	IndiaNivesh	7.37



Methodological Rigor

- I. Cross-Validation: Hyperparameters tuned using GridSearchCV with 3–5 folds.
- II. Scaling Decisions: Linear models and MLP scaled for convergence; tree-based models optionally scaled.
- III. Feature Engineering: ROCE capped, efficiency score computed, outlier treatment applied.
- IV. Model Comparison: Evaluated R^2 , RMSE, MAE across 8 models.
- V. Residual Diagnostics: Verified residuals for homoscedasticity and normality.
- VI. Interpretability:
- VII. Linear models: Coefficients interpreted for direction and magnitude.
- VIII. Tree ensembles: Feature importance extracted and compared across models.
- IX. MLP: Permutation importance used to approximate feature impact.

Recommendations & Next Steps

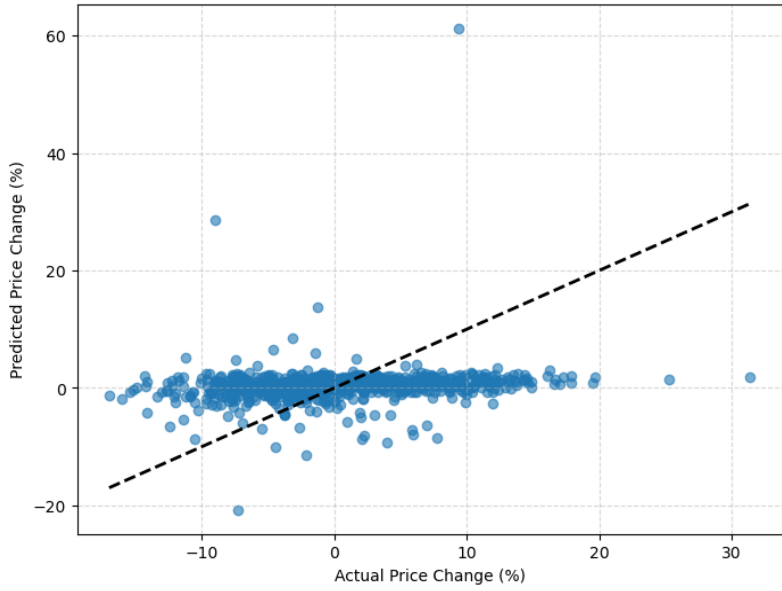
- I. Focus investments on companies with high market capitalization, ROCE, and ROA.
- II. Use Random Forest predictions to identify top stocks each quarter.
- III. Expand models with macroeconomic indicators (inflation, interest rates, global indices).
- IV. Consider time-series approaches (LSTM, RNN) for sequential prediction.
- V. Update dataset quarterly for improved model relevance and accuracy.

Conclusion

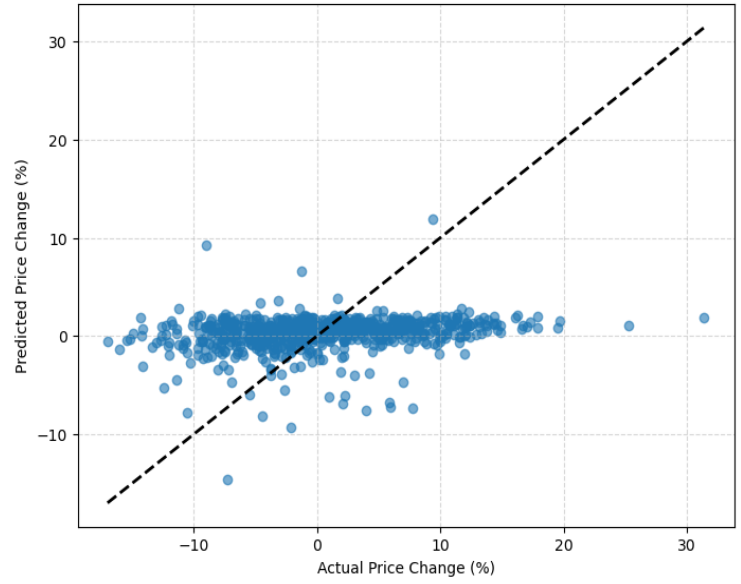
This analysis demonstrates a thorough, multi-layered approach, combining exploratory analysis, statistical modeling, machine learning, and predictive ranking. Tree-based ensembles, especially Random Forest, offer the most reliable predictions, while linear models provide interpretable insights. The analysis identifies key financial drivers, top stocks to watch, and lays groundwork for continuous investment strategy refinement.

Appendix

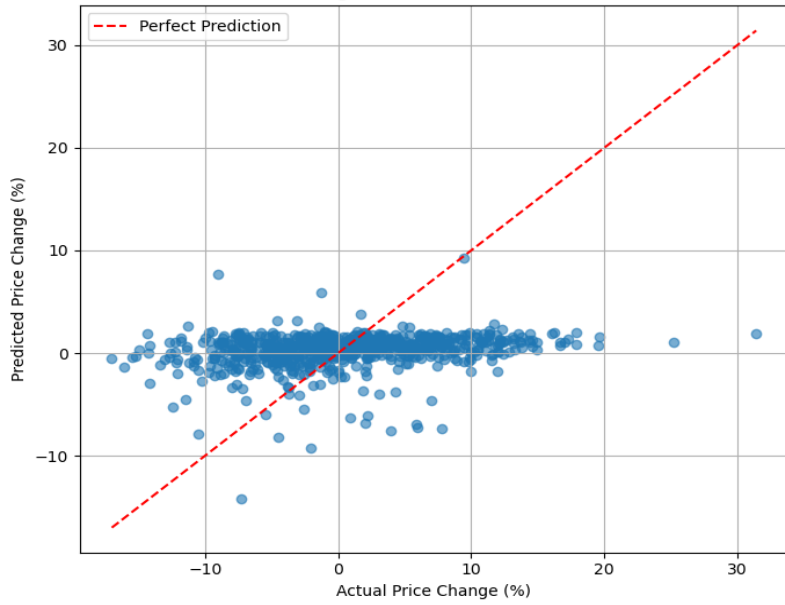
Linear Regression: Actual vs Predicted



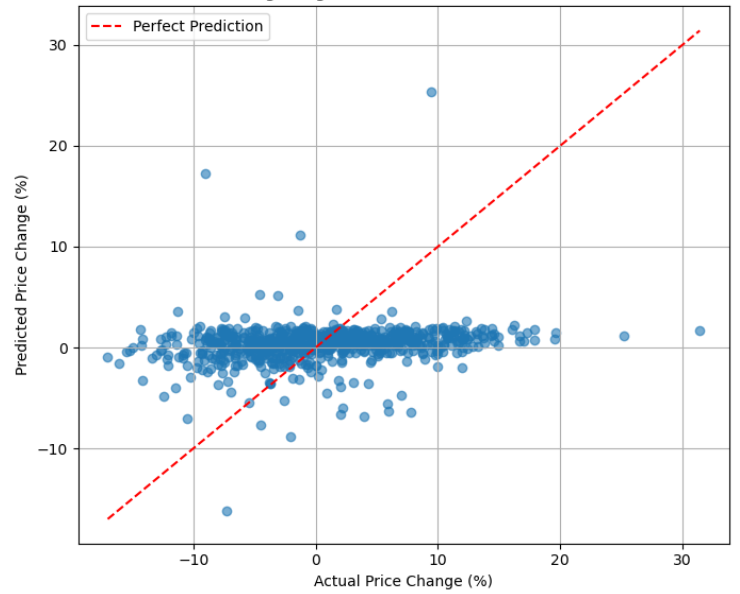
Elastic Net Regression: Actual vs Predicted



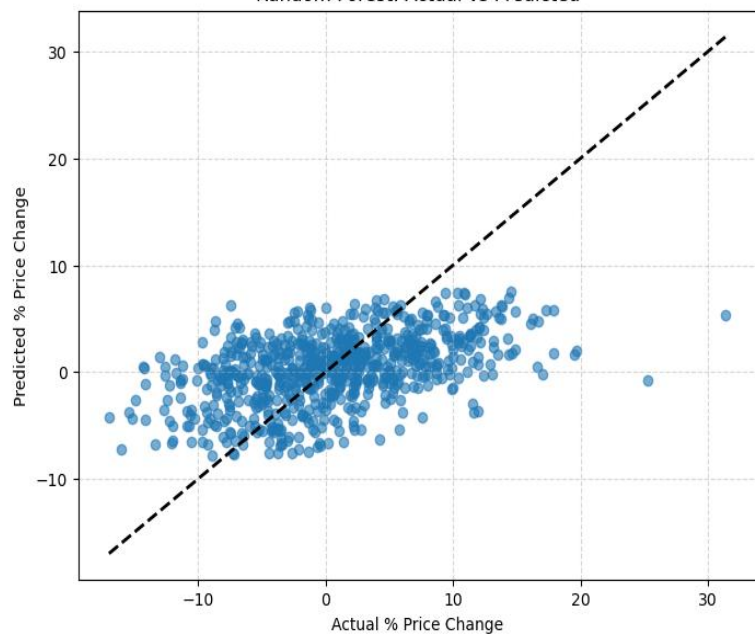
Lasso Regression: Actual vs Predicted



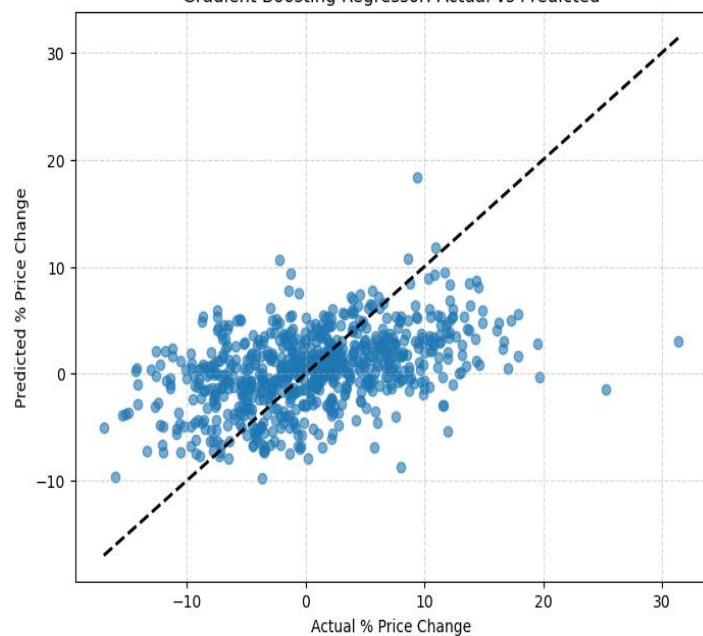
Ridge Regression: Actual vs Predicted



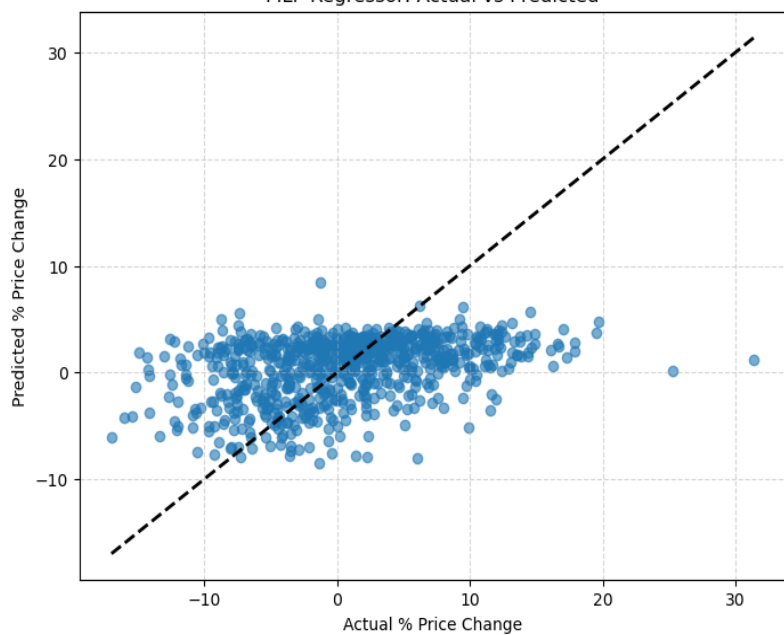
Random Forest: Actual vs Predicted



Gradient Boosting Regressor: Actual vs Predicted



MLP Regressor: Actual vs Predicted



XGBoost Regressor: Actual vs Predicted

